



Review article

How can machine learning be used in stress management: A systematic literature review of applications in workplaces and education



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ABSTRACT

In today's competitive and rival world, stress has emerged to be an integral part of every person's life which affects an individual directly or indirectly in many ways. The COVID-19 pandemic even glorified the importance and cruciality of managing stress, anxiety and depression as these created a massive impact on the economy, education, healthcare, business areas and other aspects of society in every possible manner. This study determines to find all the feasible contributing factors to stress, anxiety and depression which influence individuals coming from vivid occupational backgrounds due to personal, work-related, psychological and interpersonal reasons. Our research aims to define and describe the impact of the rise in technology and the COVID-19 pandemic on the stress levels of an individual. It includes various supervised and unsupervised machine learning algorithms in detecting stress efficiently and effectively among a huge population. The objective of this paper is to make those millions of people aware of the early detection and treatment of stress before it becomes life-threatening to them. The paper finally throws light on how stress-related research will help policymakers in the education field and general industry sector to rebuild the policies on stress and countermeasures to avoid stress.

1. Introduction

Stress is a feeling of physical or mental tension or anxiety that a person comes across in case of unavoidable circumstances or any thought which makes one restless, angry and frustrated. It is the body's reaction to any change it experiences which a person expresses in the form of physical, mental or emotional responses. In simple terms, an individual feels stressed when the expectations forecasted and reality struck are distant making one restless and tensed (Parslow et al., 2004). Stress can be positive, that is stress keeps a person active, boosted and ready for any peril. On the other hand, stress can have an overall negative impact on an individual if the stressors (viz. stress-related factors) persist for longer durations without relaxation or waiting causing a breakdown. We very well understand that stress cannot be uprooted at a stance but we can at least identify and address various factors which cause stress in people coming from different occupational areas. Therefore, getting through the stressors is very important to develop an effective stress management plan accordingly (Amanvermez et al., 2022; Kao & Cheng, 2022).

There are several reasons why stress is normally caused to humans. Some of them are financial obligations that people face in life, emotional stress, and workplace/career-related (Amanvermez et al., 2022). Students go through academic stress, while traumatized individuals face

PTSD, and post-traumatic stress. People find it difficult to cope in the face of changes in the status quo, uncertainty, loss of close ones, illness or physical health issues, etc.

Several advancements happened in past years regarding stress. Especially during the COVID-19 pandemic the amount of stress increased due to housework, online classes and assignments and covid infection (Vogel et al., 2022). There are various machine intelligent techniques available which can help in predicting stress. The LDA (Linear Discriminant Analysis) is of its kind a machine learning technique used for binary classification between a stressed condition and a non-stressed condition. Support Vector Machines, SVMs' are a great tool for classifying stress into different levels according to severity like low (manageable), medium (under concern) and high (immediate action needed) incorporating the procedure of processing signals according to the respective classification technique (AlShorman et al., 2022). The naïve Bayes algorithm is using probabilistic theory to perform the classification task. XGBoost stands for EXtreme Gradient Boosting. Gradient boosting is a machine learning technique for regression and classification problems, which produces a prediction model in the form of an ensemble of weak prediction models, typically decision trees (Gan et al., 2022). Random Forest algorithm can be intensively used to classify patients suffering from schizophrenia, a disease which impacts the reading, writing and

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thinking skills of a person making them unaware of their surroundings and the harsh reality (Rois et al., 2021; Salankar et al., 2021).

There are several deep learning approaches too that have been used in data mining and image processing applications. CNN (convolution neural network), ANN (artificial neural network) and RNN (recurrent neural network) are among the more popular techniques in a deep learning framework. CNN s' can be a great deep learning algorithm to measure psychological stress faced by college students extensively. The long short-term memory (LSTM) network is an extended structure of the recurrent neural network (RNN) and has been used in previous mood prediction problems (Gedam & Paul, 2021).

Although there are umpteen stress management techniques implemented by various organizations and universities for employees and students, It is also important to discuss the factor that causes stress. Our review systematically lists the various factors causing stress and is limited to students and employees in general. The factors are categorized into Education-related factors (Students), Education-related factors (Teacher), Workplace related factors and COVID-19 related factors, each category has been compared against the classification of factors namely, Psychological, Personal, Work-Related, and Interpersonal factors.

To summarize, the key contribution of the paper is as follows:

- i Our systematic review identifies different stress /prediction methods using machine learning and deep learning techniques.
- ii The paper lists and explains various factors that affect stress and its impact on automatic prediction of stress machines and deep learning in the education and workplace sector?
- iii The paper also discusses how the rise in technology impacted the overall increase in the stress levels of an individual.
- iv The paper also adds discussion about stress-related research during the COVID-19 pandemic.

Our systematic review on stress management focuses on the technological aspect of the human being. In other words, it does not only cover the technological side of identifying stress using intelligent techniques such as machine learning and deep learning, but it also lists various factors that affect stress, causes of stress, technological fear and its impact on stress. The paper will be a viewpoint for educationalists, policy-makers and industrialists to implement appropriate strategies and keep best practices of stress management in the universities and workplace.

The section of the paper is as follows. The section defines the concept map and the search criteria of the review. Section 3 explains various stress prediction methods, Section 4 lists various factors to be taken into the consideration during stress management. Section 5 discusses the various technological impact on causing stress, and Section 6 overviews the research on stress prediction related to COVID 19 pandemic. Section 7 concludes the review.

2. Search criteria and concept map

We began our study by listing keywords on stress management. As stated in the introduction section, we have limited our study to the education and workplace sector to avoid scope irrelevance. We found that most of the stress management techniques are based on static parameters (discussed in a later section). We also found that most stress-related problems begin with financial and work-related issues. The detailed concept map is drawn in Fig. 1.

2.1. Search criteria and research questions

Search criteria of a systematic review include searching relevant documents from Scopus, Web of Science and Google Scholar platform. We search the documents using various strings such as "stress management", "stress level + ML", "Stress detection + deep learning", "Stress management + COVID" etc. A total of 125 documents were retrieved

Table 1

Keywords, inclusion and exclusion criteria of the proposed systematic literature review.

Criteria	Description
Keywords	Stress management, stress + ML, Stress +DL, Stress+ Machine learning, +Stress +Deep learning, Stress detection+ machine learning, Stress detection + Deep learning, Stress+ COVID, Stress management + Machine learning, Stress management+ Education, Stress management + Workplace, Stress in Teaching-learning, Student stress in COVID 19 in universities, Employee stresses management+ COVID 19, Stress management+ COVID 19
Search repositories	Scopus, Web of Science Database, ERIC, Google Scholar
Journal Publisher	IEEE Transactions, Elsevier Springer, MDPI, SAGE, Taylor and Francis, Frontiers.
Total Number of documents retrieved	125 (including PhD thesis, Conference papers, and White papers)
Inclusion criteria	Relevance of the topic, Indexing (Scopus and ESCI/SSCI/SCI/SCIE, PubMed and ERIC indexed papers only), Conference papers with wither maximum citations or flagship conference, experimental papers, Information management papers
Exclusion Criteria	Low-quality papers, short papers, student papers, conference papers with low quality or not cited, white papers, PhD Thesis, Review papers on stress management, papers focusing on management aspect (non-technical papers), editorials.
A total number of documents filtered.	53

during filtering criteria Out of which 53 documents were taken into the consideration. Table 1 describes the keywords, inclusion and exclusion criteria of the proposed systematic literature review.

Our systematic survey on stress management focuses on answering the following research questions:

- RQ1: What are the different stress identification/prediction methods using machine learning/deep learning in Stress management?
- RQ2: what Factor affects stress and what is its impact on automatic prediction of stress using intelligent techniques in education and the workplace?
- RQ3: Has the rise in technology impacted the overall increase in the stress levels of an individual?
- How?
- RQ4: How COVID-19 pandemic affected the increase in stress levels of an individual? How?

In the furthermore sections, we seek to answer the research questions after analyzing various documents that we listed in Table 1.

3. Stress identification/prediction methods in stress management using machine learning/deep learning (RQ1)

Stress has become a severe cause of many diseases in modern times which directly or indirectly affects the person physically, emotionally, mentally and socially. According to the American Institute of Stress, about 33% of people feel extremely stressed which is a great proportion to note and a matter of deep concern [citation]. Therefore, it becomes crucial for us to develop or design systems, make methodologies and implement the right techniques in the right manner to detect stress at early stages to avoid this disastrous peril before it gets out of our control and reaches.

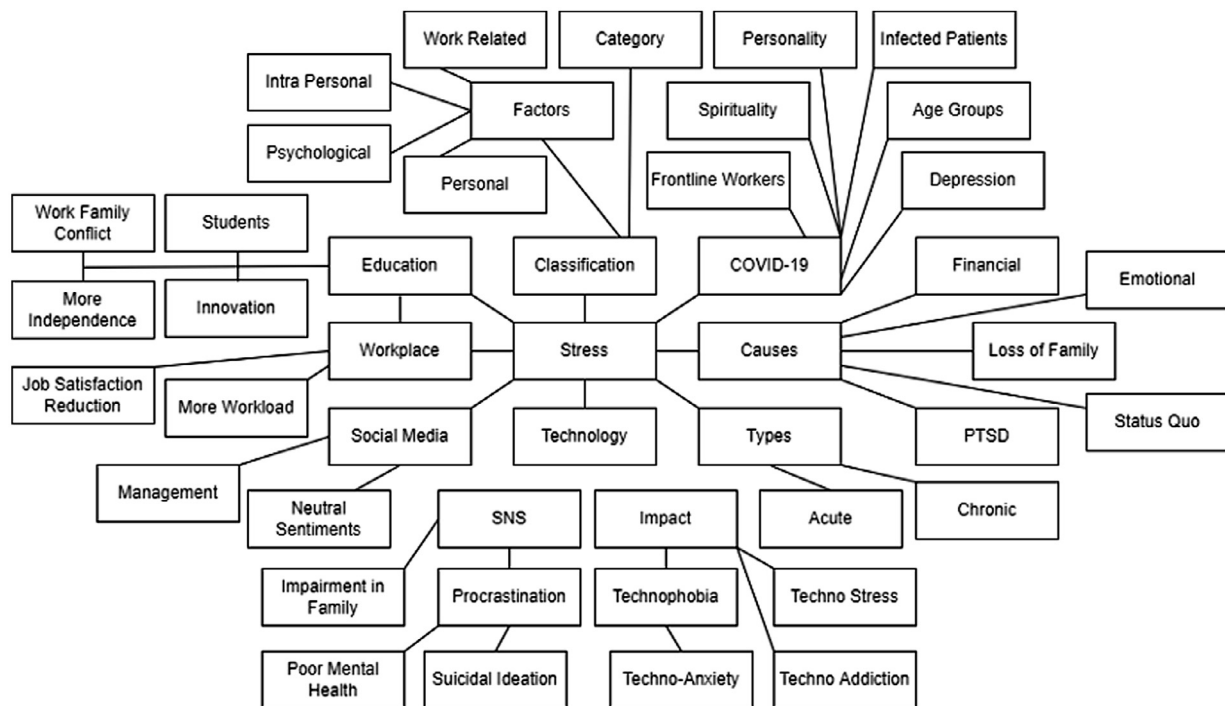


Fig. 1. Concept map of our systematic literature review.

Traditionally, stress was detected with the help of questionnaires, medical tests, rapid mock rounds or performance analysis over long time horizons focusing on an individual basis. These methods were very specific, pointed and centered around a person which in some way or the other affected the measures' objective and purpose. The primitive techniques relied on an individual's responses in the case of mock-up and questionnaire rounds and health oriented in case of medical tests which too depended on the time at which the samples were being taken making these methods influential and prone to manual errors. These methods lacked transparency and precision. Therefore, the need for modern technologies and methods arose to tackle the problem of stress effectively.

Today there exists systems, methods, techniques or algorithms which detect stress at the primary level and address this issue at wider levels that too with accuracy, consistency and precision. This section consists of descriptions of the various methodologies that can be used for stress management.

Machine learning is one of the fundamental techniques used to detect stress in recent years (either supervised or unsupervised learning). Supervised learning from these is used the most for solving mental health problems (Mutalib et al., 2021). The most commonly applies algorithms are SVM (support vector machine) followed by decision tree and then neural networks. All three algorithms can prevent overfitting, and have accuracy above 70% (Grządzielewska, 2021; Li & Liu, 2020; Mutalib et al., 2021). A decision tree is a decision-making tool that uses a tree structure for making decisions for a particular problem. It is one way to display an algorithm that contains only conditional and control statements. Decision trees are being widely used for getting to an informed conclusion about whether the person is stressed or prone to stress by analyzing various metrics in detail. They can be used to know the mental state of an individual and classify him/her as stressed even if the initial data is of subjective type. Subjective Data includes data from sources such as Electronic Health Records (EHRs'), Electronic Clinic Records (ECRs'), Interviews, Reddit and various other media which can then be pre-processed using various NLP (Natural Level Techniques) like lemma, POS (part of speech) and CUI (concept unique identifier) which identifies the key-

words related to stress like tension, depressed, mental pressure and many more to get an overall theoretical view about the stress-related words. Then the linguistic data needs to be converted to numerical data so that various classification algorithms like DT, Logistic Regression or SVM can be used to find the values (Prasanthi & Prasanthi, 2021; Tian, 2022).

The LDA (Linear Discriminant Analysis) is another machine learning algorithm that was used for binary classification between a stressed condition and a non-stressed condition (Li & Liu, 2020). The naïve Bayes algorithm is using probabilistic theory to perform the classification task. It is based on Bayes Theorem to find the probability of A is happening, given that B has occurred. Here, B is the evidence and A is the hypothesis with an assumption that the predictors/features are independent (Mutalib et al., 2021).

Logistic Regression is one of the fundamental algorithms to solve classification problems by implementing the concept of probability, but it uses a more complex cost function, which is called as Sigmoid function. Independent values (x) are combined linearly using weights or coefficient values to predict an output value (y) (Mutalib et al., 2021). Support vector machines (SVM) are a class of linear algorithms that can be used for classification, regression, density estimation, novelty detection, and other applications. SVM is using classification techniques to build a predictive model. SVM algorithm is purposely to find a hyper-plane in an N-dimensional space that distinctly classifies the data points (Liu, 2021; Mutalib et al., 2021).

SVMs' are a great tool for classifying stress into different levels according to severity like low (manageable), medium (under concern) and high (immediate action needed) incorporating the procedure of processing signals according to the respective classification technique. The classifier detected the level of stress or pressure from the naturally available data collected, such as Respiration, Finger Temperature, Skin Temperature GSR Hand, Heart Rate, and EEG, from various sources in vivid situations and at the time of driving. The data collected was then divided into different time slots such as 100, 200 and 300 s. After that, all the mathematical features were removed and fed to the separator. Finally, it was concluded that SVM classified that the pressure was reached with an accuracy of 98.41% in 100 s and 200 s concurrently and with an

accuracy of 99% in 300 s (Prasanthi & Prasanthi, 2021; Subhani et al., 2017).

XGBoost stands for Extreme Gradient Boosting. Gradient boosting is a machine learning technique for regression and classification problems, which produces a prediction model in the form of an ensemble of weak prediction models, typically decision trees. It builds the model in a stage-wise fashion as other boosting methods do, and it generalizes them by allowing optimization of an arbitrary differentiable loss function (Liu, 2021).

There are several deep learning approaches too that have been used in data mining and image processing applications. CNN (convolution neural network), ANN (artificial neural network) and RNN (recurrent neural network) are among the more popular techniques in a deep learning framework. For stress detection, deep CNN algorithms with triplets loss function have been used (Masood & Alghamdi, 2019).

The neural network is a brain model architecture with the elements of the input layer, hidden layer, connection weight, and output layer. The hidden layer is a processing layer that converts input into output. Connection weights are expressed as the relative strength of the input. It has a summation function and transformation function in each node in the input and hidden layer. The input layer consists of nodes that represent the input variables. Meanwhile, the output layer is presenting the output variable of a prediction problem (Mutalib et al., 2021).

CNN s' can be a great deep learning algorithm to measure psychological stress faced by college students extensively. To get to the desired output, the first step is to collect the ECG signals by the ECG signal acquisition system, then the wavelet transform method is used to denoise the collected ECG signal. After that, the sequential backward selection algorithm is used to select the features of psychological stress indicators to reduce the feature subspace to improve the computational efficiency and reduce error. Finally, based on the convolutional neural network, a mental pressure indicator recognition model is established and the model parameters are optimized to get the college students' mental pressure indicators (Tian, 2022). The long short-term memory (LSTM) network is an extended structure of the recurrent neural network (RNN) and has been used in previous mood prediction problems (Yu & Sano, 2020).

Each mental health problem (stress, depression, and anxiety) is tested in different models by each algorithm with a feature selection process and based on the results it can be found that the number of students with extremely severe stress and depression is less but students with extremely severe anxiety is more than 100 students, therefore, it is assumed that students are unfamiliar with about their future (Mutalib et al., 2021). A decision tree is classified as the best model for stress detection (Grządzielewska, 2021) with six attributes positive feelings, memory, negative feeling, personal relationships, leisure and religion (Mutalib et al., 2021).

When we train generalized machine learning models from human data, a common problem is that the distributions of the data may vary from one participant to another due to individual differences. Two participants with the same level of well-being levels may have different mobile phone usage habits, which causes internal covariate shifts among datasets and inequality in feature representation. Therefore, we propose an LSTM structure that uses batch normalization (BN) to reduce the heterogeneity among participants (Li & Liu, 2020; Yu & Sano, 2020).

Convolutional neural networks (CNN), which can effectively capture the spatial and temporal dependencies in an image through the application of filters (convolutional kernels), are actively used in deep learning, especially in the computer vision application area. Combining the discussions above, CNN can extract features with convolutional kernels and RNN can leverage timestamp dependencies. Since our data is in a high-dimension feature space, the training models have higher risks to be over-fitted. With the convolutional kernels on CNN, we can actively extract high-level features from our high-dimension data (Liu, 2021; Yu & Sano, 2020).

The deep LSTM model outperformed the CNN-LSTM model in participant-independent wellbeing prediction (Yu & Sano, 2020). From SVM, DNN, CNN and XGBoost algorithms, CNN gave the best accuracy in predicting the stress (Liu, 2021).

There exists multiple machine learning algorithm for predicting stress but the basic implementation of all the techniques remains the same. The output of each distinct algorithm applied may be dissimilar in terms of format or may be non-identical in differentiating among levels of stress but the significance of the end result is unchanged for all cases. Therefore, following steps are executed for detecting stress while using multiple machine learning algorithms:

Step-1: Input Stress Dataset: In the first phase, stress data is collected from various available sources like rapid mock test rounds, questionnaires sets and performance related information. Attributes and class labels are created on the dataset based on the explicit requirements of the user. If a student is under consideration then; study pressure, teachers' behavior, school environment etc. are important factors to be taken but in case of working professionals' (Malik et al., 2022) these attributes are somewhat irrelevant. So, only the contributing factors are to be incorporated.

Step-2: Pre-Processing Data: In the second phase, data is checked for any unnecessary fields or attributes which do not form an essential part of stress detection process. The data is also tested for missing values, duplicate values and null values too. The tuples containing any type of the mentioned values will be removed from the dataset so as to avoid any illogical mistakes while processing the dataset for prediction.

Step-3: Train-Test Split: In the third phase, dataset is divided into training and testing data. Mostly, 70% will be split as training data and the rest 30% as testing data. Training data is one through which the model will learn to differentiate among stress levels and find threshold values for distinct situations. The trained model will be implemented on the testing data to check for overfitting or underfitting problems.

Step-4: Implementation of Stress Detection Model: In the fourth phase, the desired algorithm will be applied for detecting stress levels based on vivid attributes or features inputted depending on the type of user under observation.

Step-5: Performance Evaluation: In the last phase, the model performance will be evaluated in terms of various parameters like accuracy, precision, F-score etc. to check the correctness of the algorithm used.

Hence, the general architecture of implementing algorithms for detecting stress is similar in all cases.

There are various datasets used in predicting/identifying stress, few datasets are open access. List of few datasets are listed in Table 2.

4. Factors affecting stress and its impact on automatic prediction of stress using intelligent techniques (RQ2)

This section covers various factors which affect stress prediction and its impact on the automatic prediction of stress using intelligent techniques. We have categorized the factors into general, education-related factors (student and teacher), work-related factors, COVID related factors. All the categories which are defined are overlapping viz. there might be some factors which might appear in multiple categories. Furthermore, the stress factors are also divided into personal, work-related, psychological and interpersonal factors. Table 3 has a matrix which lists the factors based on various papers that we cited.

4.1. General category

The stressors included in this category are the ones that lead to stress to people directly or indirectly independent of the occupational back-

Table 2
Various dataset used in stress management research.

Dataset Name	Contributors	Link
College students Stress management data	statistics tutor (statisticsanswered.com)	https://data.world/sigmastudent/college-students-stress-management-data
Student Stress Survey Jan2020 OPENDATA	Vivien Rolfe, Figshare	10.6084/m9.figshare.11559528.v1
Implications of Tamarkoz Method of Meditation on Stress, Emotion, Spirituality and Heart Rate	Nasim Bahadorani; Jerry W. Lee; Leslie R. Martin (La Sierra University LLUSD Continuing Dental Education)	https://www.openicpsr.org/openicpsr/project/137,281/version/V2/view;jsessionid=EDBCE2E5498C29916B3062D4C1BAFD4B
Effects of testimonials on the acceptance of digital stress management trainings among university students: a randomized controlled trial	Jennifer Apolinário-Hagen	10.7802/2287
Can we learn to manage stress? A randomized controlled trial carried out on university students	Dalia Saleh; Nathalie Camart; Fouad Sbeira; Lucia Romo, PLOS ONE	10.1371/journal.pone.0200997
Most common causes of stress at work in the United Kingdom in 2020	Statista	https://www.statista.com/statistics/1,134,031/causes-of-stress-at-work-in-the-uk/
Share of workers with higher, equal or lower stress levels since COVID-19 2020	Statista	https://www.statista.com/statistics/1,169,836/covid-stress-level-of-workers-in-select-countries/
Data from: What teachers in a high-performing high school need to effectively manage workplace stress: an evaluation study	DataCite (University of Southern California Digital Library (USC.DL))	10.25549/usctheses-c40-442,934

grounds from which they come. In other words, the factors included here are common to all individuals and have the same impact on them.

- a **Personal Factors:** These factors include age, gender, education, race, occupation and nationality which are major drivers of stress. Such factors affect the individual's performance to an extent and also hamper their morals. Though these factors have no genetic races but still they help in shaping human beings. These factors contribute to social exclusion, violence and discrimination among various groups of people.
- b **Work-Related Factors:** It entails any work or work environment or work issue events that affect a working professional's throughput and productivity. Some of the factors include job role, job status, experience and personal-work balance. These factors play an important role in job satisfaction. When these functions as a motivator, they result in creativity and satisfaction and consequently removes

boredom. But when they function negatively, they lead to low job satisfaction and aggression.

- c **Psychological Factors:** These relate to an individual's attitude, motive and belief toward his life. Psychological stressors depict how a person will react or respond when faced with an uncertain event or episode which in turn is the cause of stress. Two of the most vital factors are state anxiety and depression. These factors lead people to shrink their network and isolate themselves up and stop doing lots of activities they enjoy.
- d **Interpersonal Factors:** These incorporate the relationship among people i.e. the kind of relationship that one maintains with peers and colleagues in offices or say teachers and students in schools or universities. These include emotional and mental harassment or mistreatment and also the troubles caused by their family members, friends and seniors. Taking these factors in a positive way, having good peers can help increase ones performance, whereas, having peers who push others in a negative situation lead to reduction in ones performance.

4.2. Education-related factors (Students)

Stress is not new but yes stress among students is somewhat a new area of discussion. Today, students from the inception of their secondary education (sometimes during primary education too) are burdened with loads of work say homework and assignments which is one major reason for stress among students. Therefore, identifying and addressing academic stressors at the right time in the right way is very important.

- a **Personal Factors:** These factors affect an individual's inner self and compel it to overthink leading to stress. Some examples include physical appearance, internet addiction, low confidence level and self-desire. Such factors make one conscious of itself. Factors such as physical appearance, self-desire, etc. are not that important. If one makes themselves happier rather than having a perfect body is more beneficial. If one falls into the well of such factors, leads to low self-confidence and self-esteem.
- b **Work-Related Factors:** The environment and atmosphere in which a student studies and completes their learning process is very important to consider as they are the building blocks of the student's future. Some factors which influence a student's academic performance include Study facilities, competitive environment, peers' behavior and the teacher-student relationship. Factors such as highly competitive environment, good teacher-student relationship, etc., if taken in a positive way, motivates students to their best and not just good enough. Whereas, if these factors are taken on the wrong side, it makes the student more aggressive, low confident, etc.
- c **Psychological Factors:** These factors depict how a student will react or respond to scheduled events like examinations, viva-voce and practical tests and unscheduled events like random mock tests or exams. Accounting for such factors is as important as they may affect a student's performance adversely. Some examples include Exam anxiety, fear of failure, low grades, and tough learning.
- d **Interpersonal Factors:** These relate to relationships a student has with his teachers, friends and other related entities around him as they influence a student's behavior and working style which may in turn cause stress in some future time. Factors include crowded classrooms, bullying by friends and members and team performance. These factors have direct impact on the environment of a student and leads to negativity at the workplace. We need students who can appreciate others and motivate them from time to time.

4.3. Education-related factors (Teachers)

Teachers are those sources of knowledge who enlighten the lives of many students for the sake of their brighter future. They provide us guidance, moral values, etiquette and advice so that all of us can

Table 3
Various categories of stress and stress factors.

Category/Stress Factors	Personal	Work-Related	Psychological	Interpersonal
General	Age, Gender, Race, Education, Occupation, Nationality	Job Status, Job Role, Experience, Personal-Work balance	State Anxiety, Depression	Interpersonal Differences, Troubles caused by peers'
Education-related factors (Students)	Physical appearance, Internet addiction, Low confidence level, Self-Desire	Study facilities, competitive environment, peers' behavior, teacher-student relationship	Exam anxiety, fear of failure, low grades, tough learning	Crowded classrooms, bullying by friends and members, team performance
Education-related factors (Faculty)	Self-esteem, attitude, supervision abilities, problem resolver	Long working hours, job insecurity, depressing work mood, deviant behavior	Delayed material provision, emotional exhaustion	Emotional intelligence, efficient interaction, team performance
Workplace related factors	Rank/designation, personal inadequacy, self-conflict	Workload volume, work relationships, control over workload, bad coordination among members	Stress, anxiety, depression	Interpersonal differences, more tasks assigned
COVID-19 related factors	Marital status, household responsibilities, lack of personal time	Abuse, sexual harassment, gender discrimination, insufficient opportunities	Fear of infection, risk of community transmission, friends and family emotional support	Interpersonal differences, working with and for peers'
Citation	Babore et al. (2020), Carroll (2020), Giusti (2020), Kroth (2019), Labrague & de los Santos (2021), Naveed & Aslam (2017); Ramón-Arbués (2020); Saeed et al. (2020); Shammi, et al. (2020); Wani (2021)	Babore (2020), Carroll (2020), Giusti (2020), Kroth (2019), Labrague & de los Santos (2021), Naveed & Aslam, (2017), Ramón-Arbués (2020), Reddy et al. (2018), Saeed (2020); Shammi (2020); Soto, (2021); Wani (2021), Zhou et al. (2021)	Babore (2020); Giusti (2020); Labrague & de los Santos(2021), Naveed & Aslam (2017), Palattao, Solano et al. (2021a), Reddy (2018), Shammi (2020); Wani (2021); Zhou (2021)	Babore (2020); Labrague & de los Santos (2021), Naveed & Aslam (2017), Ramón-Arbués (2020), Reddy (2018), Saeed (2020); Soto(2021); Zhou (2021)

be successful. Though they give us their everything yet we as students sometimes fail to be compliant with our teachers or say teachers if not by us but due to other factors may be in stress. Therefore, understanding and taking appropriate steps at the right time is very important.

- a **Personal Factors:** These factors let teachers undergraduate their performance due to excessive thinking and wrongly interpreting their skills or calibre. Some examples include Self-esteem, attitude, supervision abilities, and problem resolution. Teaching process can be affected positively or negatively based on teachers' personality. If a has a negative personality, it will to a negative environment of the classroom which in turn will cause negativity in student learning. On the other hand, if a teacher has a positive personality, it will lead to increased outcome for both the students and the teachers.
- b **Work-Related Factors:** These factors incorporate workplace-related physical, mental and emotionally traumatic events which cause stress among teachers. Some of the Factors are long working hours, job insecurity, depressing work mood, and deviant behavior. These

factors leads to reduced productivity and efficiency of the teachers which in turn leads to decline in job satisfaction, team unity, increased absenteeism, etc.

- c **Psychological Factors:** These factors relate to teachers' attitudes and responsive behavior towards known events or unknown turned-out stressful situations. These factors include Delayed material provision and emotional exhaustion. High level of these factors is leading to poor health and well being of teachers causing teachers burnout, lack of engagement, highest turnover rates, etc.
- d **Interpersonal Factors:** It entails factors related to teachers' behavior, their surrounding pupils' attitude or nature (i.e. say students), social factors and openness to discussions and team activities. Some factors are Emotional intelligence, efficient interaction and team performance. These factors have direct impact on the organization culture. These factors leads nowhere and in turn spoils the work environment. We need more people who can understand and motivate others from time to time.

4.4. Workplace related factors

Offices are one of the most stressful places to work and survive. In corporate offices and workplaces, stress is not just a factor of workload or work pressure but also dependent upon environmental, social and idiosyncratic factors. Hence, defining and describing them is crucial.

- a **Personal Factors:** These factors relate to an employee's or working-class professional's self-profile. Such factors are dependent upon the individual completely and influence their productivity and performance towards the company adversely if not considered prior. Examples include Rank/designation, personal inadequacy, and self-conflict. When employees face such factors, it makes them hard to concentrate, make decisions and feel confident in what they are doing. It also leads to absenteeism, emotional disorders and work-life imbalance.
- b **Work-Related Factors:** These factors majorly relate to the individual's working environment and work-related problems or issues which they face during their employment tenure. Examples include Workload volume, work relationships, control over workload and bad coordination among members. These factors make employees more prone to errors, poor work performance, mental health problems, burnouts and conflicts in the workplace.
- c **Psychological Factors:** These explain the responsiveness and perceptions of the working professional towards official as well as unofficial events or situations. Some factors include depression, anxiety and stress. Psychological disorders lead employees to depression and anxiety, sleeping disorders which in turn causes drop in the work performance and satisfaction.
- d **Interpersonal Factors:** These factors relate to the employee's relationship with their peers, employer and other teammates or group members while assigned group projects. Also, these factors incorporate social and work-related behavior and attitude of seniors and their colleagues. Examples include Interpersonal differences and more tasks assigned. Such factors have direct impact on the work life and balance and working environment. Outside the workplace it can be considered good as it helps workers to manage the stress caused by workplace stressors.

4.5. COVID-19-related factors (Frontline workers)

COVID-19 had affected and is still affecting the whole world in a way that was never imagined by anyone but in this tough time, our doctors and nurses were saviours for the entire nation as they were working day and night as frontline workers that too efficiently and effectively. Nurses were mostly stressed during those last 1–2 years period because of long working hours and adverse working conditions. Therefore, it became necessary to discuss those stressors to overcome any such situation in future.

- a **Personal Factors:** These factors relate to the nurses' personal life, personality, family background and job status (i.e. full-time or partially working) as it affects their professional life to an extent also. Some examples include Marital status, household responsibilities and lack of personal time. Many frontline workers faced such factors during the pandemic. Many took such factors in a positive way to help as many patients and they could whereas some, who took it in a negative way, lead to decrease in their performance and satisfaction.
- b **Work-Related Factors:** These factors depict how work culture, work hours, work problems or work issues, working environment and work overload can affect their performance and job efficiency. Some factors are abuse, sexual harassment, gender discrimination and insufficient opportunities. During the pandemic, such factors were very common for the frontline workers to face which led to decrease in the work they were doing and the work satisfaction.

IMPACT OF INCREASED USE OF TECHNOLOGY

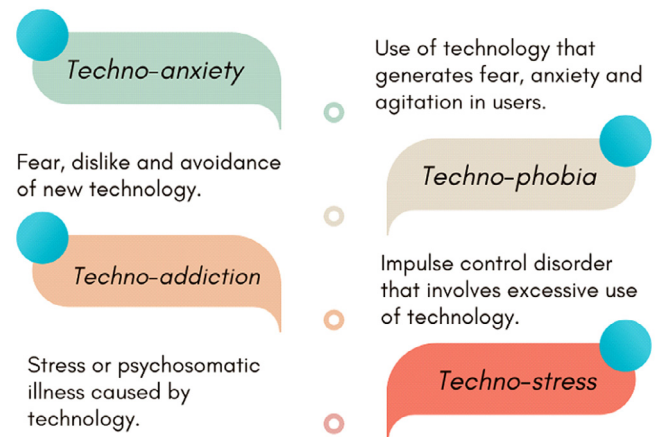


Fig. 2. Types of stress in technology use.

- c **Psychological Factors:** It entails the factors which explain how mentally or emotionally nurses would respond to tough or critical situations which can later turn out to be severely stressful as the conditions were also not normal at that very time. Some examples include Fear of infection, risk of community transmission and support from family and friends. These factors led to sleeping disorders to the frontline workers, increased anxiety and depression and reduced work-life balance for the frontline workers.
- d **Interpersonal Factors:** These factors explain the relationship of nurses with their peers, colleagues and coworkers. These also demonstrate teamwork efficiency and case handling capacity too. Some factors include interpersonal differences and working with peers. These factors helped the frontline workers to overcome the stress caused by the workplace stressors. But these factors had a direct impact on the environment and the peer groups.

5. Has the rise in technology impacted the overall increase in the stress levels of an individual?

Modern communication technologies have made our lives easier, exchanges that would take weeks to happen before can now happen within minutes or seconds on the other side of the world. But they have also increased the rate at which they move. Modern technology has gained such a presence in our lives that it's easy to become addicted to its use to the point it starts impacting our lives in many ways. This research would look at how technology has impacted our lives and contributed to an increase in an individual's overall stress levels. First of all, users of information and communication technology aka ICT have been documented to experience stress, worry, and tension as a consequence of the technology use (See Fig. 2). This is a condition known as techno-anxiety (la Torre et al., 2019). Secondly, users can also experience psychological effects which could lead to decreased confidence levels. Such circumstances can lead to emotions of helplessness and discomfort, as well as an aversion to or fear of using computers, a condition known as technophobia (la Torre et al., 2019). Thirdly, using ICTs excessively can lead to a condition known as techno-addiction (la Torre et al., 2019). And lastly, Technostress is a condition of the modern times of adaptation due to the inability of coping with new computing technology in a productive outcome and healthy manner (Dong et al., 2020).

SNS aka Social Networking Sites/Services have a huge impact on causing the above-mentioned conditions in individuals nowadays. They are challenging self-control on numerous levels for an increasing num-

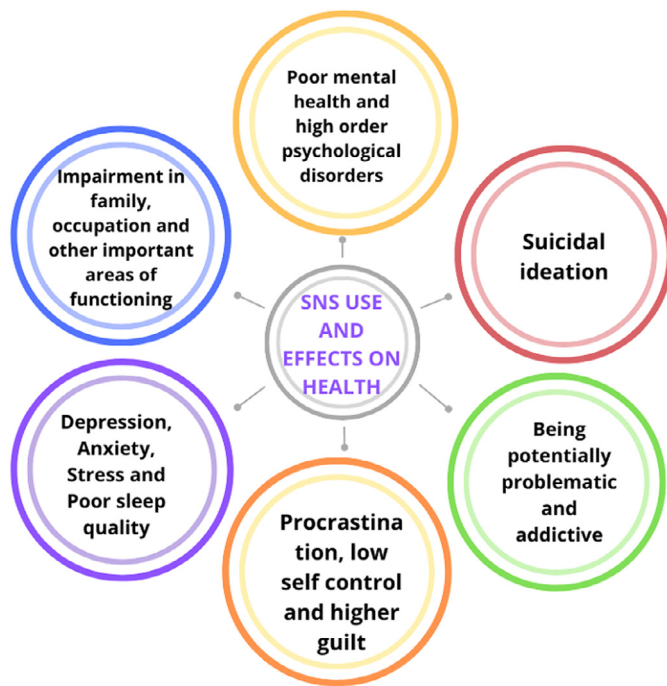


Fig. 3. SNS use and effect on health.

ber of people joining the internet from different spheres and aspects of life. (Meier et al., 2016). SNS is a platform where people can share their life happenings in the form of photos, videos, etc. and react to other people's posts to build a network or a relationship with like-minded people around the world (Hussain & Griffiths, 2019). Increased use of SNS can incorporate increased use of ICT devices such as smartphones, desktops, tablets, etc. (Meier et al., 2016). There is evidence that excessive usage of social media sites (Chauhan & Palivela, 2021) i.e., more than two hours per day is linked to poor mental health, increased psychological distress, and suicide thoughts, as well as becoming potentially harmful and addictive for a tiny number of people (Hussain & Griffiths, 2019; Kuss et al., 2018). For such people, the terms like 'SNS addiction' or 'problematic SNS use' (PSNSU), have been rightfully termed as an engaging activity that occupies an individual's thoughts. It is motivated by a strong (and sometimes unhealthy) desire to log on or use SNSs. The individual thus ends up spending a lot of time and effort on SNSs. This has a gravely negative impact on other social activities, interpersonal relationships, work/studies, and/or psychological health and well-being (Hussain & Griffiths, 2019; Kuss et al., 2018). Furthermore, more studies showed that psychiatric disorder variables like OCD, anxiety, ADHD, depression and stress as a result of SNS addiction (Hussain & Griffiths, 2019).

Sleep problems were also a significant health problem that has been associated with technology use (Hussain & Griffiths, 2019). (See Fig. 3)

Additionally, a few researchers have looked at the predictors of procrastination through the usage of the internet and technology (Meier et al., 2016). 'Procrastination' is defined by Meier et al. (2016) as the "failure of self-regulation of not exerting enough self-control for task management". Notably, the use of SNS that can provide short-term guilty pleasures by the experiences and the enjoyment of certain activities such as gambling, pornography, etc. as compared to important but unpleasant academic tasks, has led to low trait self-control and decreased self-directed learning in children implying frequent uncontrolled and possibly procrastinatory use of SNS (Kuss et al., 2018; Meier et al., 2016). As a result, to sum up, we may say that disrupted internet use causes severe distress and/or problems in crucial areas of personal, educational, familial, social, occupational, and other functioning (Király et al., 2020).

Since early 2020, the world has now been facing the covid-19 pandemic. Several emergency measures were imposed by the state (see Fig. 4). These included the procedures to access remote working from home by both public and private organizations (Király et al., 2020; Molino et al., 2020). The government aimed to reduce the gathering of people and limit the movement of people throughout the national territory without compromising the services offered by them (Király et al., 2020; Molino et al., 2020). This step taken by the government influenced every profession. Though workers were able to adapt to the drastic change all of a sudden due to the rapid development of ICTs. It expected workers to respond to it positively and rapidly but yet there was a fear of technology (Alhumaid et al., 2021). In the educational sector, the fear was about inadequate preparedness and technical readiness (Alhumaid et al., 2021). In the medical and health sector, the patients were constantly worried about any results that suggested serious conditions or wrong diagnoses (Alhumaid et al., 2021). Whereas in the banking sector, the fear emerged due to the attitude of customers toward technology. Customers are worried about mobile payments and providing their data (Alhumaid et al., 2021). Finally, concerning households, less attention has been paid to the use of technology because of the fear of technology and the increased number of family responsibilities (Alhumaid et al., 2021; Chakraborty & Kar, 2021).

Considering the workload that has been increased, nowadays teachers are expected to incorporate the use of technology into their teaching methodologies in the classrooms (Dong et al., 2020). This creates excess workload, challenge and stress for teachers. Teachers are constantly struggling with the time available and the increasing demands of the schools and colleges. Keeping pace with new emerging technologies and the innovations associated with them has been very stressful and challenging for the teachers. Even if teachers are seeing technology as a tool for knowledge delivery, lesson preparation, or a way to attract students, they lack the required adequate skills and competencies for implementing this constructive use of technology (Dong et al., 2020).

On the contrary, organizations have increased their investments in ICTs (Hwang & Cha, 2018). By doing so, benefits for both companies and workers have been found but along with them, certain detriments have all been reported. Changing the internal structure of the companies also changes the traditional workplace stress to mobile workplace stress including increasing employee engagement to fulfill their job responsibilities, improving personal performance, and thus achieving organizational goals (Wang et al., 2021). Such conditions which lead to techno-stress are also associated with adverse psychological outcomes such as increased level of role stress, reduced job satisfaction and reduced organizational commitment. Also, a bad information system (IS) results in decreased innovation in employee tasks, reduced productivity when using the IS and dissatisfaction with the IS (Abid et al., 2019; Hwang & Cha, 2018).

Starting from kindergarten until leaving high school, children almost spend at least 15,000 h in classrooms and the experience they are having there is one of the most crucial predictors of their well-being and who they'll become in the later stages of their lives (Oberle & Schonert-Reichl, 2016). Positive and supportive classrooms can help students thrive, and become successful, whereas, a stressful classroom can jeopardize the healthy development of a child and success in school (Oberle & Schonert-Reichl, 2016). Teachers play a central role in maintaining the environment of a classroom and therefore teachers' burnout levels (stress that is not maintained well) affect the environment of a classroom. More burnout will the stress students (Oberle & Schonert-Reichl, 2016). During the current pandemic, learners are experiencing a non-conductive emotional state to focus on learning, which can also impact their self-directed learning process and health (Shirish et al., 2021).

This is not only true for school students but also undergraduate children. Out of 3670 medical students, 93% turns out to own a smartphone while 83% had a laptop or desktop computer. Under given conditions, only 1505 students (i.e. 41%) considered themselves to be physically and mentally capable of engaging in online learning (Baticulon et al.,

COVID-19 And Technology use Impact

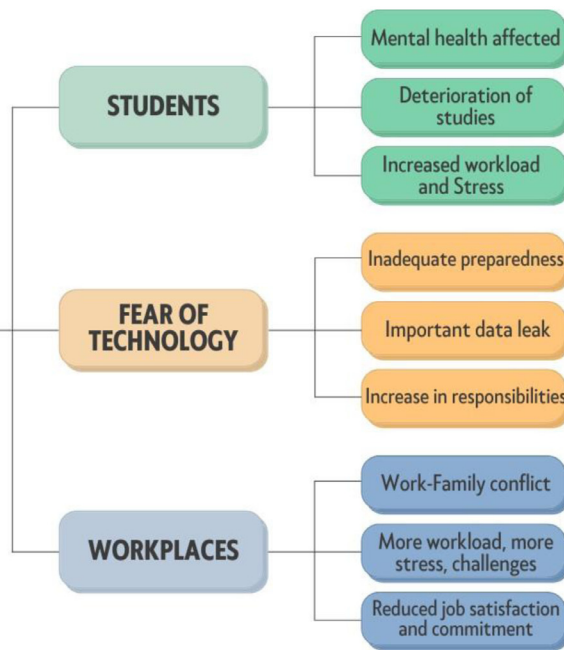


Fig. 4. Stress factors and COVID 19.

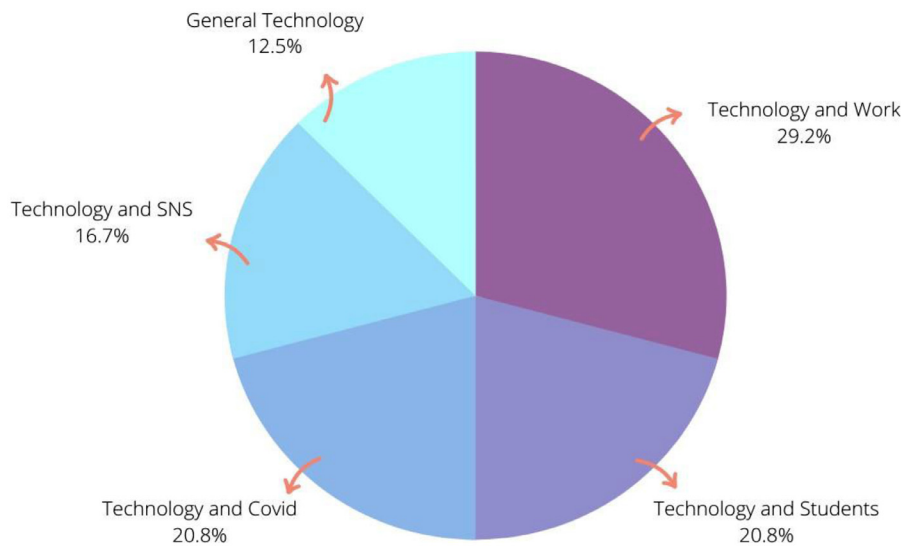


Fig. 5. Stress-related research work in various domains.

2021). To access other online resources, 79% of the students had a post-paid internet subscription while only 19% used prepaid mobile data. The most frequently encountered barrier out of all, was the difficulty in adjusting learning styles, having to perform responsibilities at home, and poor communication or lack of clear directions from educators. All these barriers increased their workload thereafter (Baticulon et al., 2021).

As observed, significant factors in predicting intention to use mobile learning were perceived fear and expectation confirmation. Previous studies moreover, showed how the use of mobile learning (ML) in the field of education, amid the pandemic, offered a potential outcome for teaching and learning. This impact may be reduced by the fear of losing friends, a stressful family environment and fear of future results in school. Therefore, during and also after the pandemic, it is important to examine students appropriately to enable them to handle the situation both emotionally and mentally (Alhumaid et al., 2021). (See Fig. 5)

6. How COVID-19 pandemic affected the increase in stress levels of an individual?

The Coronavirus disease, also known as COVID-19, was declared a pandemic on 11th March 2020 by the World Health Organization (WHO). It is an infectious disease that is caused by a novel coronavirus, called the SARS-CoV-2 virus. This pandemic led to a worldwide lockdown, shutting down every aspect of our regular and day-to-day lives. The ongoing consequences and the aftereffects of this pandemic have caused significant amounts of stress not just to adults but also to the younger generations of mankind (Yasmin, 2020). Fig. 6 depicts the flow of discussion by listing the major keywords about stress and COVID 19 related research.

With everyone locked inside their homes, the number of hours spent by us in the virtual zone drastically increased. The pandemic saw people resort to more focus on their online/virtual lives by a multifold. With

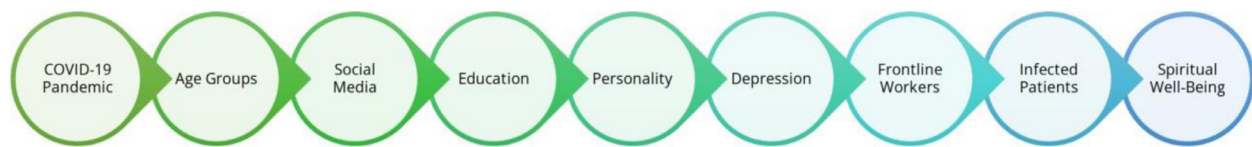


Fig. 6. Flow of discussion COVID 19 and Stress.

mental health issues rising and stress levels peaking, a large number of people started talking and commenting online about the stress, anxiety and trauma that resulted from the pandemic. Speaking about the Indian virtual scenario, results show how death and the lockdown were the primary aspects that caused stress and mental health issues (Kar et al., 2021). Though it was also observed that the same audiences online that were active on Twitter, posted tweets with neutral sentiments. Studies furthermore have highlighted the fact that a large section of people had a neutral sentiment in a broader picture. It can also be concluded that the Indian audiences have increased the use of social media to fill up the extra free hours the lockdown provided them with (Praveen et al., 2021) (Mohamed Ridhwan & Hargreaves, 2021).

The Indian government considered the pandemic a notified disaster and declared a nationwide lockdown on 24th March 2020. Several students throughout the country were amidst their examinations, including board examinations. One of the sectors that took the largest hit throughout the pandemic was the education sector. The fact that all schools, colleges and universities were closed was academically disruptive on several levels. This drastic change in the status quo proved to raise stress and anxiousness levels among all students (Khattar et al., 2020). The lockdown kept adding to more troubles as educational institutions struggled to provide the classroom environment virtually. No student had ever learned in such environments before and all had to quickly adapt to the new format. Many found it extremely tedious to study from home in virtual environments and copying in exams increased on a massive scale. It was also observed that while the students were under pressure, this scenario harmed the social life and general well-being of students as well. Even educators tried their best in this time to accommodate the needs of all students and came up with various methodologies to teach and create a stress-free environment for students. However, one more issue for students that led to more stress was the increased amount of workload in the form of extra term work, assignments, presentations, activities, etc. While this was an attempt to help students utilize the extra time on their hands better, it ended up creating very stressful environments for students throughout (Parthiban et al., 2021).

Social distancing measures and lockdowns have not only affected the older adults but also the younger ones. The pandemic forced all the schools, colleges, and educational institutions to shift their emphasis from traditional face-to-face learning to online learning environments like Microsoft Teams, Google Classrooms, Zoom, etc. While some institutions maintained their academic continuity, others lacked due to budgetary and technical constraints (Shirish et al., 2021). Though the unavoidable situations benefited the students in many ways like students got time for self-learning and avoiding the traditional method of spoon-feeding, it also negatively affected them like the enhanced use of online learning tools, there is a greater need for more learner-centric responsibility for effective learning, but all students may not perceive this increased responsibility (Shirish et al., 2021). (See Fig. 7)

Working on computers has always been challenging and tiring but it can also promote and support employee engagement and thus overall promote innovative behavior (Abid et al., 2019; Wang et al., 2021). Many disputes have also been reported like all the changes have blurred the boundaries between work and family and caused work-family conflicts which later on affect employees' innovative behavior (Santuzzi & Barber, 2018; Wang et al., 2021).

The personality of a particular person plays a significant role in how they react to situations. When faced with conditions as brutal as the pandemic and lockdown, different people with different personality types reacted to the changes and increased stress differently (Parslow et al., 2004). Studies have shown that spending long hours at home, unprecedented time on social media, self-introspection of people, dire need of information w.r.t. Vaccines and medications, deterioration of physical health, etc., led to elevated levels of depression, exhaustion, anxiety and stress. During the pandemic, simple fever and cough would lead people to get stressed and overanalyze all symptoms of the virus (Palattao et al., 2021b). Moreover, as mentioned before, different personality traits like goal striving, neuroticism, extroversion activity, conscientiousness and sociability perceived stress in their manner (Liu et al., 2021).

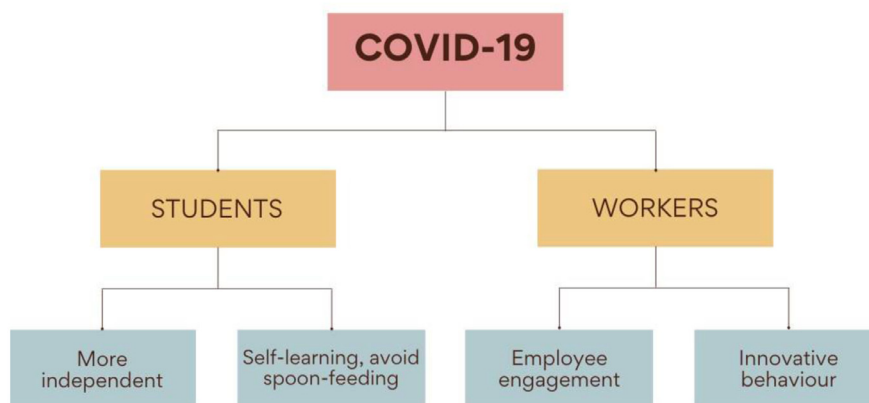
Loss of hope is how a lot of people define depression. Depression in turn drains down a person drastically. During the pandemic, all the front-line workers, doctors, nurses, scientists, etc. were a ray of hope to all. But we must consider that the amount of pressure and stress on these rays of hope would have been tremendously high, especially during peak pandemic waves (Havaei et al., 2021). Studies have shown what factors of the pandemic did wear down our frontline workers. One of the primary factors in predicting the decline in a worker's mental health is what role the particular individual has. The roles of different workers varied from the Emergency Room Staff to Nurses to Surgeons. Next, how many hours of sleep the worker got has been equally troublesome. One another aspect that we would like to point out is how much amount of news related to COVID-19 the individual worker takes in a day. Finally, one aspect that studies showed is the usage and consumption of alcohol and cannabis as one of the reasons why frontline workers might feel more stress than usual (Rezapour & Hansen, 2021). Now that we have discussed the factors that increased stress for frontline workers, we should also take a look at what could have and what did decrease stress levels simultaneously. All medical work environments were in some way or other associated with environments where the mental health of the nurses and doctors was positive. Managing workload, balance, and psychological protection were important predictors of anxiety, depression, stress, and emotional exhaustion (Havaei et al., 2021).

We have seen the effect of the pandemic on the stress levels of normal people, the education sector, social media, personality, and front-line workers. A study was conducted where the stress levels of infected patients were predicted by asking micro-EMA questions. Using the micro-EMA questions, we can predict from a range of days how the stress levels are fluctuating in infected patients (Li et al., 2022).

The spiritual well-being of individuals is widely assumed to affect the stress management of a person to some extent. A recent study with sociodemographic characteristics was conducted that has highlighted some interesting observations. The stress COVID-19 Scale, Paloutzian & Ellison Spiritual Well-Being Scale, the Adult Hope Scale and the Resilience Scale were used to collect data. Contrary to popular belief, this study concluded that the spiritual well-being of an individual cannot in any way predict the stress of COVID-19. Moreover, individuals with strong spiritual selves need not have lower stress levels. Along with spiritual well-being, even resilience and hope cannot be considered predictors of stress during this pandemic (Nooripour et al., 2021).

Positive Effects of

Fig. 7. Positive effects of COVID 19.



7. Discussion and conclusion

7.1. Contributions to the literature

Stress seems to be an abstract word each time when we listen about it but today this term has great significance in lives of many people either directly or indirectly irrespective of their personal, professional, social and demographic factors which can affect people adversely in case, if not anticipated at an early stage. At present, there exists multiple ML models to predict stress and even more advanced methodologies are being implemented to make stress detection more reliable. Therefore, it is explicitly understandable that how crucial stress management is in this era of competition. We attempt to give an overview of the contribution of discussed ML models and techniques in real life. Some of the applications in which stress detection can be of importance are:

- Stress detection models can be used to predict stress in school and college students at prior stages with the help of stress scaling scores on the basis of a set of questions related to the activities performed by students.
- Working professionals spend most of their time in offices and have to often engage themselves in vivid team projects or tasks at various instances too where in the competition of winning over others led them to tension like situations causing them to overthink and affect their health also. Multiple models are being integrated to systems to keep a track of each and every action performed by employees at different times of a day when in office so as to get an overall index. Few examples were given in many information management related journals viz. (Adamovic, 2022) explores stress related research for telework employees about impact of cultural perspective, (Rezvani & Khosravi, 2019) explores stress issues in software industry employees, (Dwivedi et al., 2020) explores education, work, life related factors, (Wang et al., 2021) discusses role stressors, job satisfaction related test cases and proposes meditation model, (D'Arcy & Teh, 2019) predicted the security related stress among the information security employees. (LeRouge et al., 2006) investigated impact of role stress fit and self-esteem factors of IT professionals. In near future, the outcome of our systematic literature review can be used in examining stress of an employees working individual sector such as healthcare, IT industry, Logistics Management, Nursing, tourism etc.
- In-fact, emphasis on stress prediction is being given as an input to healthcare sector as anxiety or tension is a major cause of cardiovascular and neurological problems. Stress detection monitoring systems are embedded into health checking machines so as to measure stress (Sciagrà et al., 2021).

7.2. Implications

In today's time, especially ever since the technological age has boomed up, the impact that the pandemic has had brought everything to a standstill. Every sector took the hit and it is safe to conclude, not just by inference but also by research studies and findings stated above, that stress levels of individuals reached new heights all of a sudden. With an unexpected turn of events, people from all occupations and backgrounds found themselves at home trying to keep the work going.

The major aim of the detailed review was to focus on each occupational background from which the individual is coming so that the best stressors which are affecting the particular class of people can be analyzed and according to them the accurate techniques and methods can be applied to get a proper result and output. To examine and understand how this was caused and how can this stress be rightfully managed.

The focus was on assessing a wide range of populations altogether based on the most impactful factors that cause stress among them keeping accuracy, integrity and consistency intact. Therefore, we got through various machine learning algorithms for understanding the process of finding the best features which are a great contributor to stress in people according to different demographical, social, educational and professional attributes. Mathematical and statistical results have been a major advantage of these techniques as a fixed criterion set at the initial stage of training may get along for long periods for detecting stress levels with utmost proficiency.

The rise in technology as stated above effectively helped life make easier. Though it was easier to work and coordinate and educate through technological means, it did play its role in increasing stress levels for students and at the workplace. Speaking about social media, the younger generation was already engrossed in it, but now they were joined by everyone staying at home during the pandemic. It was also observed how the overall sentiment was neutral on social media (Sharma et al., 2022; Stone & Can, 2021).

7.3. Future directions

The paper focuses on the education and industry sectors as the amount of stress is more in these fields. While searching on platforms like google scholar, Scopus, ERIC and Web of Science the maximum documents were retrieved on stress in said two sectors. Our systematic review on stress management focuses on the techno-psychological aspect of the human being. In other words, it does not only cover the technological side of identifying stress using intelligent techniques such as machine learning and deep learning but it also lists various factors that affect stress, causes of stress, technological fear and its impact on stress.

The paper will be a viewpoint for educationalists, policy-makers and industrialists to implement appropriate strategies and keep best practices of stress management in the universities and workplace. In near future, the same discussion can also be extended to other sectors like the financial or healthcare sector.

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Declaration of Competing Interest

The authors of this research study declare that there is NO conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jjimei.2022.100110](https://doi.org/10.1016/j.jjimei.2022.100110).

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